Algebra 1 Honors
Unit 10 Assessment

Name____________________
Date_________________
Period__________

1. The function $G(x) = |0.05x - 0.85| + 1.77$ models the gas price, in dollars, for January 6, 2020 to August 24, 2020, where x is the number of weeks since January 6, 2020.

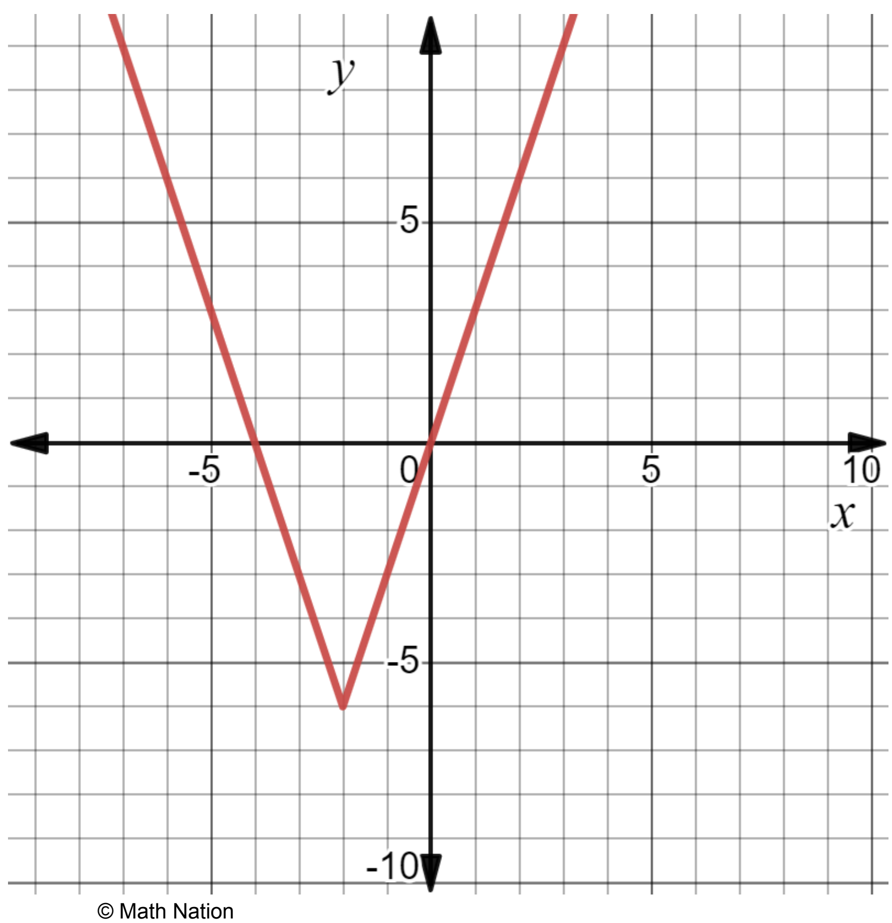
Determine when gas prices were $\$2.52$.

work	
Number of weeks since January 6, 2020	2 and 32

2. An equation is shown.

$$y = 3|x + 2| - 6$$

Use the coordinate grid to graph the equation.



Determine the solution for each equation. Show your work.

3. $\lvert 6-4k\rvert-30=0$	4. $\lvert 8x-11\rvert+45=6$
$\boldsymbol{k}=$ -6 and 9	$\boldsymbol{x}=$ No solutions
5. $\lvert n+6\rvert=19$	6. $-3\lvert y+5\rvert+27=0$
$\boldsymbol{n}=$ 13 and -25	$\boldsymbol{y}=$ 4 and -14

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7. The circumference of a regulation discus used in the men's Olympic track and field events should be 691.15 millimeters (mm). The manufacturing tolerance is within 3.15 mm. Write an absolute value equation that could be used to determine the maximum and minimum value of x, the circumference of a regulation discus used in the men's Olympic track and field events.

Equation	$|x - 691.15| = 3.15$

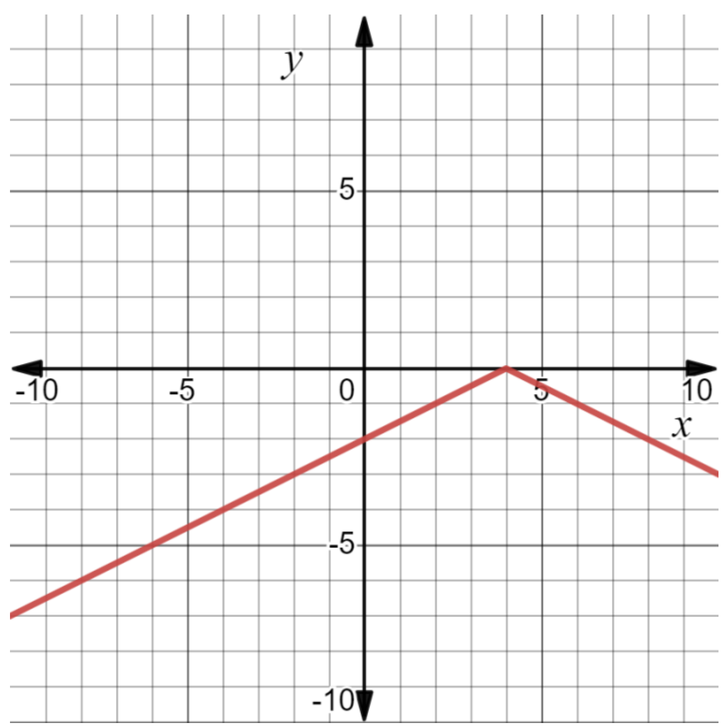
8. A table of values for the function $f(x) = -\frac{1}{2}|x - 4|$ is shown.

x	-6	-3	-2	0	3	5	6	9
y	-5	-3.5	-3	-2	-0.5	-0.5	-1	-2.5

a. Complete the table.

Name	Key Feature	Name	Key Feature
x-intercepts	$(4, 0)$	**increasing**	$x < 4$
y-intercept	$(0, -2)$	**decreasing**	$x > 4$
vertex	$(4, 0)$	**positive**	none
axis of symmetry	$x = 4$	**negative**	$x < 4$ or $x > 4$
domain	$\mathbb{R}$	**range**	$-\infty < y \leq 0$

b. Use the coordinate grid to graph the equation.



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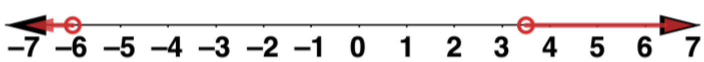
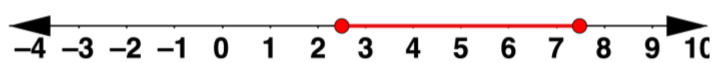
Solve each inequality. Then, graph the solution set on the number line.

9. $\lvert 4x + 5\rvert - 7 > 12$	10. $-2\lvert 5 - x\rvert + 4 \geq -1$
$x > 3.5 \qquad x < -6$	$2.5 \leq x \leq 7.5$
	

11. The U.S. Bureau of Labor and Statistics tracks the prices of everyday items. During the years 2012 to 2021, the median price of a dozen eggs was $\$2.09$. During that time the price fluctuated but was always within $\$0.88$ of the median price.

a. Write an absolute value inequality that represents x, the price of a dozen eggs during the years of 2012 and 2021.

Inequality	$\lvert x - 2.09\rvert \leq 0.88$

b. Solve the inequality. Use the blank number line to graph the range of prices of a dozen of eggs.

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